Towards Fluid–Structure Interaction Modelling of Vocal–Ventricular Fold Coupling in ArtiSynth Using the Lattice Boltzmann Method

Junheng Chan

Division of Linguistics and Multilingual Studies, Nanyang Technological University, Singapore

junheng001@e.ntu.edu.sg

Introduction. Vocal–ventricular fold coupling (VVFC) refers to mechanical coupling between the vocal and ventricular folds in the larynx. Articulatory phonetic studies suggest that interaction between these structures contributes to sounds and gestures involving glottal constriction (Esling & Harris, 2005; Moisik et al., 2011). For instance, the ventricular folds appear to "sit" on the vocal folds during the production of a glottal stop, introducing additional damping to the vocal fold oscillations. Moisik & Esling (2014) modelled VVFC with low-dimensional lumped masses and one-dimensional (1D) aeroacoustics, finding effects including damping, altered fundamental frequency and irregular vibration. Moisik & Gick (2017) subsequently developed an anatomically detailed three-dimensional (3D) ArtiSynth model of laryngeal biomechanics, but without aerodynamics or acoustics. Yong (2020) developed the existing 3D vocal–ventricular fold finite-element model (FEM) and achieved sustained oscillation using a simplified aerodynamic forcing function rather than a spatially resolved airflow field. More recently, Kummerländer et al. (2026) demonstrated two-way fluid–structure interaction (FSI) and self-sustained oscillation by coupling a 3D homogenised lattice Boltzmann method (LBM) airflow solver to a reduced-order six-mass vocal-fold model (without ventricular folds). The present study takes a preliminary step towards a continuum vocal–ventricular FSI model by incorporating a two-dimensional (2D) D2Q9 LBM airflow solver into Yong's 3D ArtiSynth FEM, using coronal sections of the fold geometry to define the fluid boundaries.

Methods. The long-term objective is two-way 3D FSI. Following an initial static integration based on Kummerländer's (2018) compute-shader prototype, an instrumented D2Q9 BGK solver was linked to Yong's ArtiSynth FEM (Lloyd et al., 2012). Three 314 × 402 lattices at 0.05 mm spacing used mesh-derived boundaries from coronal sections of the FEM. The initial 0.98 mm glottal gap spanned approximately 20 cells. Pressure patterns were normalised to a chosen 100 Pa loading scale and transferred to FEM nodes using surface-triangle areas and normals, with gravity switched off. Airflow was then recomputed on the recorded deformed geometry, and the surface loads were updated.

Preliminary results. The LBM planes were successfully aligned with the FEM geometry. The mesh-derived boundaries constrained airflow to the modelled laryngeal airway and prevented flow around the outer ventricular-fold surfaces. A central jet was produced through the open glottis and ventricular-fold gap (Figure 1). Pressure loading produced peak nodal displacements of approximately 0.084 mm in the vocal folds and 0.010 mm in the ventricular folds. Recomputing airflow on the deformed geometry changed peak jet speed by 1.65–2.96% across the three slices at matching step counts. Updated resultant pressure loads increased by 2.3–2.4% on the vocal folds and decreased by 4.9–5.0% on the ventricular folds.

Discussion. The staged LBM–FEM integration demonstrates pressure-induced deformation and geometry-dependent airflow and loading but does not constitute continuous moving-boundary FSI. The airflow remains developing and the pressure-loading scale is not physically calibrated. Further work requires stable, validated dynamic boundary and force exchange, physical calibration, and extension to a 3D lattice such as D3Q19. This provides a proof of concept towards an FSI method for investigating VVFC.

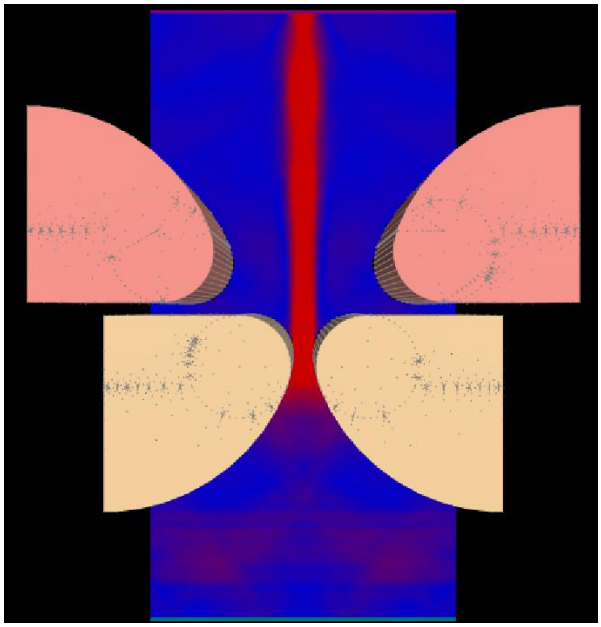


Figure 1. Initial static integration of the 2D D2Q9 LBM velocity field with the mid-coronal geometry of the 3D vocal–ventricular FEM. Colours represent relative velocity magnitude from lower (blue) to higher (red) velocity.

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